Forecasting for Economics and Business

Lecture 10: Forecasting Inter-related Series

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Economic variables are inter-related

Economic variables are often (and usually) inter-related; for example,

- when incomes increase, people consume more;
- when interest rates increase, people invest less.

We can model these relationships by regressing one variable on the other(s).

Autoregressive distributed lag (ARDL)

Because of possible delays in adjustment to a new equilibrium, we typically need to include lagged values in the regression.

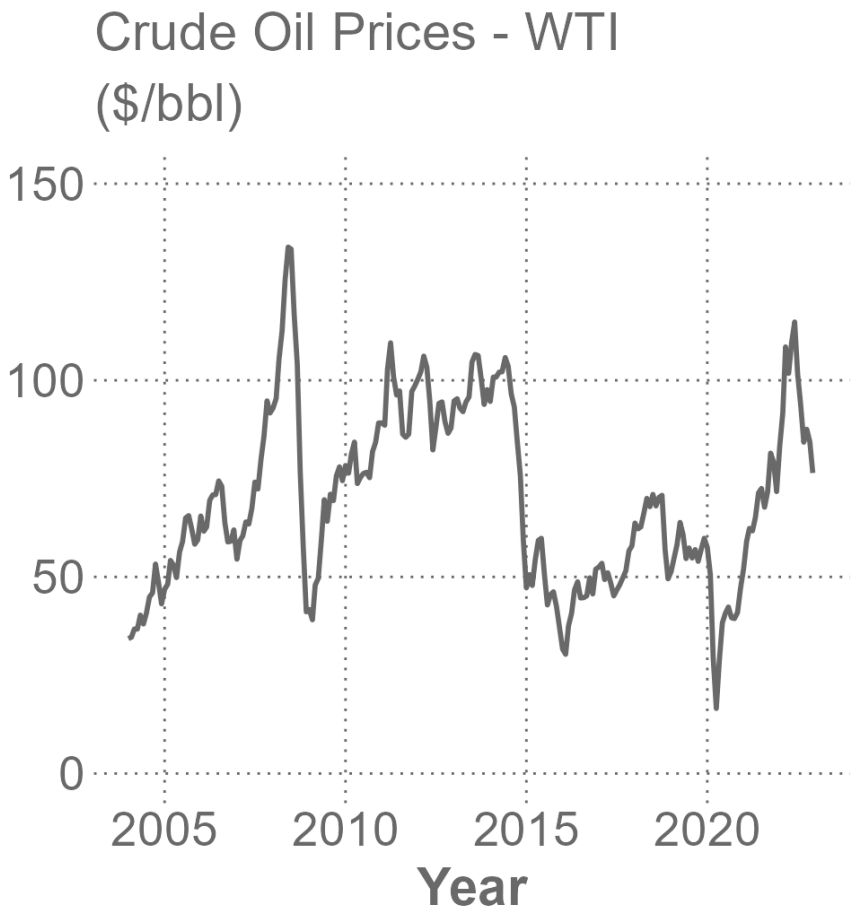
This leads to an autoregressive distributed lag (ARDL) model, as follows:

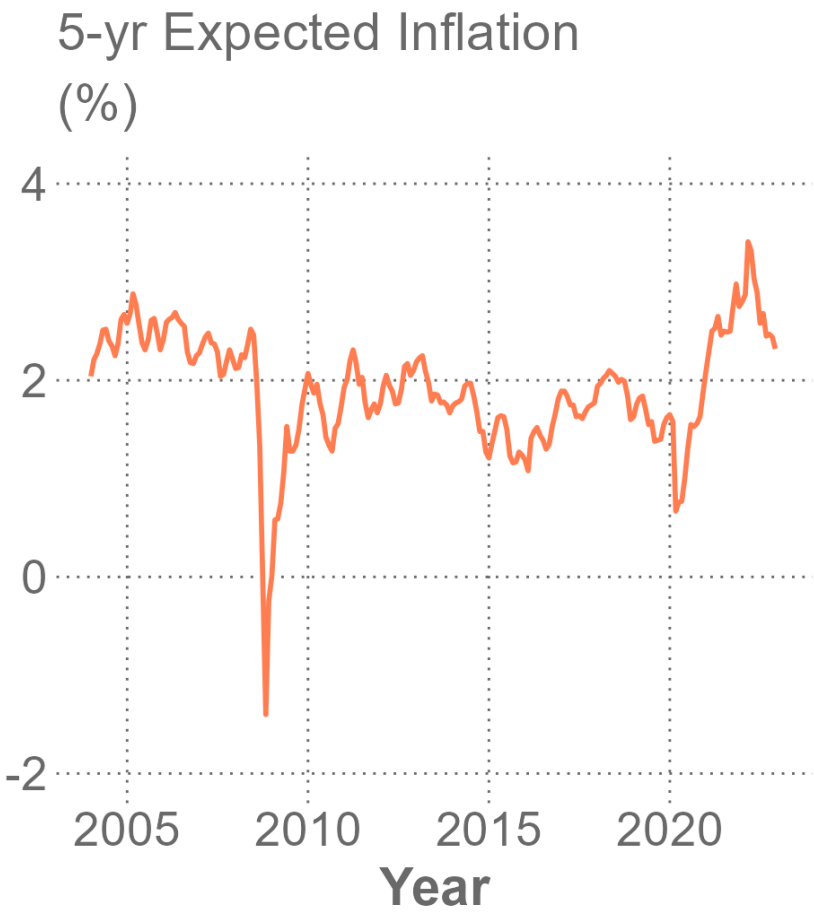
$$y_t = \alpha + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \delta_0 x_t + \ldots + \delta_q x_{t-q} + \varepsilon_t$$

Note that an autoregressive model and a distributed lag model are special cases of the ARDL.

As usual, we can use an information criterion to select p and q.

A relationship between oil prices and inflation





ARDL lag length and estimated parameters

Obtain AIC and SIC for lags 1 through 4

AIC	SIC
1.997	2.058
1.932	2.023
1.946	2.068
1.913	2.065

AIC suggests four lags while SIC indicates two lags. Let's go with SIC.

	`$\alpha$`	`$\beta_1$`	`$\beta_2$`	`$\delta_0$`	`$\delta_1$`	`$\delta_2$`
estimate	0.251	0.986	-0.098	0.016	-0.010	-0.007
s.e.	0.048	0.065	0.063	0.002	0.003	0.002

Forecasting with ARDL

Point forecasts:

$$
\begin{aligned}
y_{t+1|t} &= \alpha + \beta_1 y_t + \beta_2 y_{t-1} + \delta_0 \hat{x}_{t+1} + \delta_1 x_t + \delta_2 x_{t-1} \\
y_{t+2|t} &= \alpha + \beta_1 y_{t+1|t} + \beta_2 y_t + \delta_0 \hat{x}_{t+2} + \delta_1 \hat{x}_{t+1} + \delta_2 x_t \\
y_{t+h|t} &= \alpha + \beta_1 y_{t+h-1|t} + \beta_2 y_{t+h-2|t} \\
&\quad + \delta_0 \hat{x}_{t+h} + \delta_1 \hat{x}_{t+h-1} + \delta_2 \hat{x}_{t+h-2}
\end{aligned}
$$

Note, and this is one of the caveats of using ARDL in forecasting, we need to have a knowledge of future realized values of x. Often, that involves also (separately) forecasting those values, or setting them to some values (e.g., scenario forecasting).

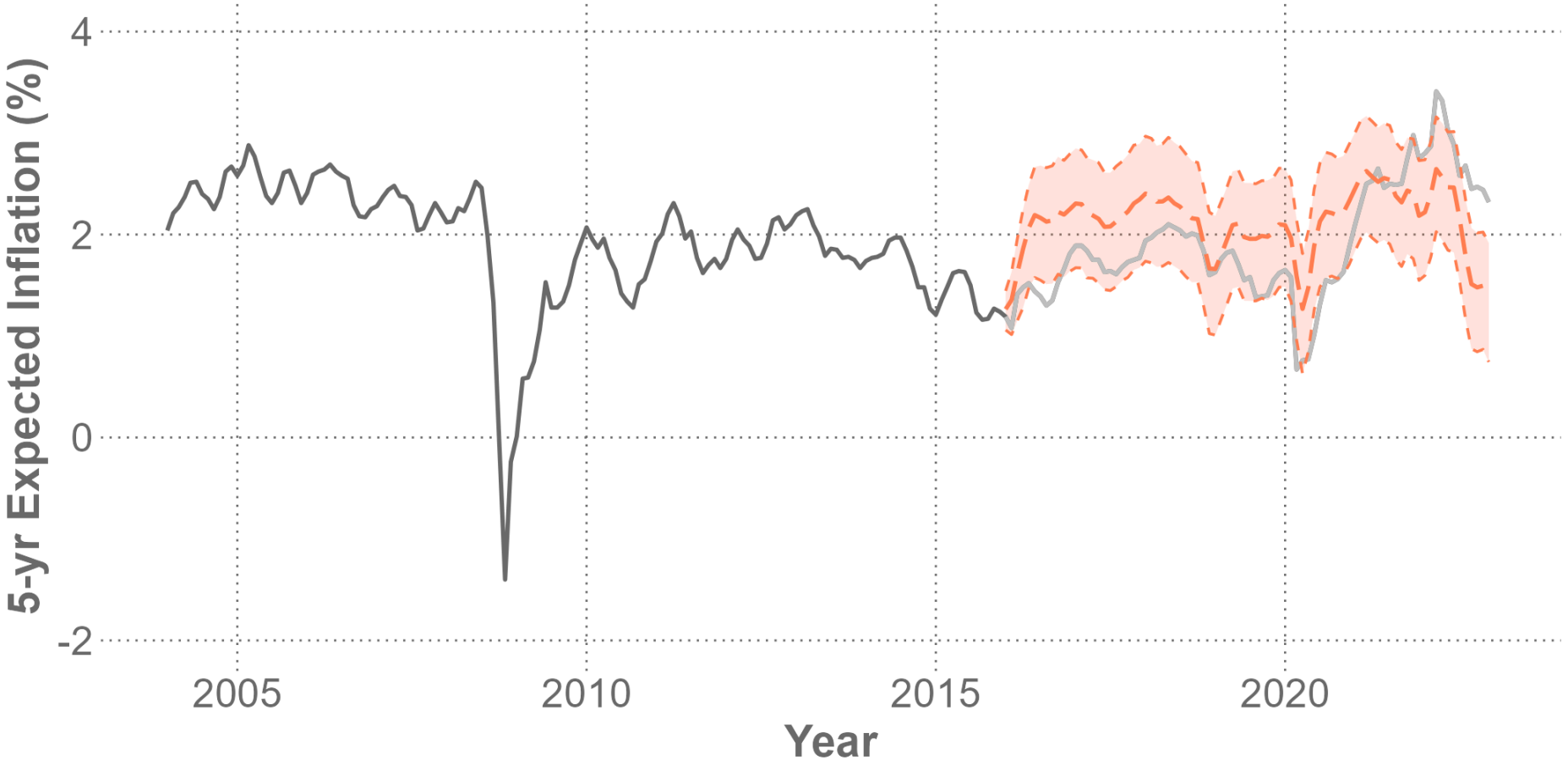
Forecasting with ARDL

If we assume (somewhat unrealistically) that we know the future realized values of $\boldsymbol{x}$, then the forecast errors and forecast error variances from the ARDL model will be the same as those from an AR model.

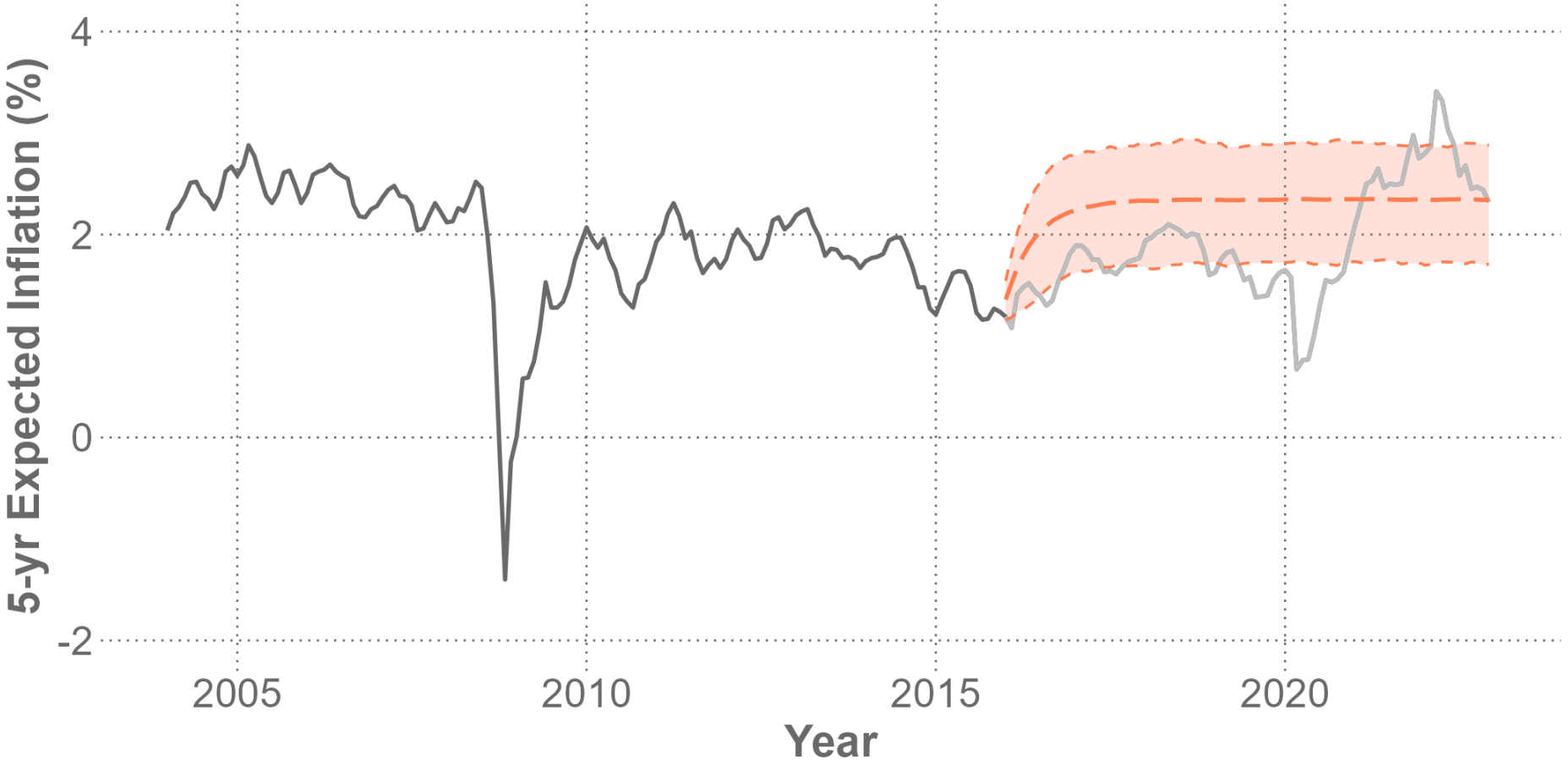
This is, in a way, akin to working with an AR model that includes a trend or a seasonal component.

Of course, obtaining direct multistep forecasts is always an option.

If we knew future realized values of oil prices



If we set future realized values to the current oil price



Vector autoregression

More generally, dynamic linkages between two (or more) economic variables can be modeled as a *system of equations*, better known as a vector autoregression (VAR).

Vector autoregression

To begin, consider a bivariate first-order VAR.

Let $\{X_{1t}\}$ and $\{X_{2t}\}$ be the stationary stochastic processes. A bivariate VAR(1), is then given by:

$$x_{1t} = \alpha_1 + \pi_{111} x_{1t-1} + \pi_{121} x_{2t-1} + \varepsilon_{1t}$$
$$x_{2t} = \alpha_2 + \pi_{211} x_{1t-1} + \pi_{221} x_{2t-1} + \varepsilon_{2t}$$

where $\varepsilon_{1t} \sim iid(0, \sigma_1^2)$ and $\varepsilon_{2t} \sim iid(0, \sigma_2^2)$, and the two can be correlated, i.e., $Cov(\varepsilon_{1t}, \varepsilon_{2t}) \neq 0$.

Vector autoregression

An n-dimensional VAR of order p, VAR(p), presented in matrix notation:

$$\mathbf{x}_t = \alpha + \mathbf{\Pi}_1 \mathbf{x}_{t-1} + \ldots + \mathbf{\Pi}_p \mathbf{x}_{t-p} + \varepsilon_t,$$

where $\mathbf{x}_t = (x_{1t}, \ldots, x_{nt})'$ is a vector of n (potentially) related variables; $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{nt})'$ is a vector of error terms, such that $\mathbb{E}(\varepsilon_t) = \mathbf{0}$, $\mathbb{E}(\varepsilon_t \varepsilon_t') = \Sigma_\varepsilon$, and $\mathbb{E}\left(\varepsilon_t \varepsilon'_{s \neq t}\right) = 0$.

A parameter matrix of a vector autoregression

$\Pi_1, \ldots, \Pi_p$ are n-dimensional parameter matrices such that:

$$\Pi_j = \begin{bmatrix} \pi_{11j} & \pi_{12j} & \cdots & \pi_{1nj} \\ \pi_{21j} & \pi_{22j} & \cdots & \pi_{2nj} \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{n1j} & \pi_{n2j} & \cdots & \pi_{nnj} \end{bmatrix}, \quad j = 1, \ldots, p$$

Features of vector autoregression

General features of a (reduced-form) vector autoregression are that:

- only the lagged (i.e., no contemporaneous) values of the dependent variables are on the right-hand-side of the equations.
  - Although, trends and seasonal variables can also be included.
- each equation has the same set of right-hand-side variables.
  - However, it is possible to impose different lag structure across the equations, especially when p is relatively large. This is because the number of parameters increases very quickly with the number of lags or the number of variables in the system.
- the autregressive order, p, is the largest lag across all equations.

Modeling vector autoregression

The autoregressive order, p, can be determined using system-wide information criteria:

$$AIC = \ln|\Sigma_\varepsilon| + \frac{2}{T}(pn^2 + n)$$
$$SIC = \ln|\Sigma_\varepsilon| + \frac{\ln T}{T}(pn^2 + n)$$

where $|\Sigma_\varepsilon|$ is the determinant of the residual covariance matrix; n is the number of equations, and T is the total number of observations.

Estimating vector autoregression

When each equation of VAR has the same regressors, the OLS can be applied to each equation individually to estimate the regression parameters - i.e., the estimation can be carried out on the equation-by-equation basis.

Indeed, taken separately, each equation is just an ARDL (albeit without any contemporaneous regressors in the right-hand side of the equation).

When processes are covariance-stationarity, conventional t-tests and F-tests are applicable for hypotheses testing.

Testing in-sample Granger causality

Consider a bivariate VAR(p):

$$
\begin{aligned}
x_{1t} &= \alpha_1 + \pi_{111} x_{1t-1} + \cdots + \pi_{11p} x_{1t-p} \\
&\quad + \pi_{121} x_{2t-1} + \cdots + \pi_{12p} x_{2t-p} + \varepsilon_{1t} \\
x_{2t} &= \alpha_1 + \pi_{211} x_{1t-1} + \cdots + \pi_{21p} x_{1t-p} \\
&\quad + \pi_{221} x_{2t-1} + \cdots + \pi_{22p} x_{2t-p} + \varepsilon_{2t}
\end{aligned}
$$

- $\{X_2\}$ does not Granger cause $\{X_1\}$ if $\pi_{121} = \cdots = \pi_{12p} = 0$
- $\{X_1\}$ does not Granger cause $\{X_2\}$ if $\pi_{211} = \cdots = \pi_{21p} = 0$

Testing in-sample Granger causality

In our example:

$\textbf{Crude Oil} \xrightarrow{gc} \textbf{Inflation}$: F statistic is 12.12

$\textbf{Inflation} \xrightarrow{gc} \textbf{Crude Oil}$: F statistic is 1.98

So, we have a unidirectional Granger causality (from crude oil prices to 5-year expected inflation).

Forecasting with VAR models: one-step ahead

To keep things simple, we illustrate using bivariate VAR(1).

Point forecasts:

$$x_{1t+1|t} = \alpha_1 + \pi_{111} x_{1t} + \pi_{121} x_{2t}$$
$$x_{2t+1|t} = \alpha_2 + \pi_{211} x_{1t} + \pi_{221} x_{2t}$$

Forecast errors:

$$e_{1t+1} = x_{1t+1} - x_{1t+1|t} = \varepsilon_{1t+1}$$
$$e_{2t+1} = x_{2t+1} - x_{2t+1|t} = \varepsilon_{2t+1}$$

Forecasting with VAR models: one-step ahead

Forecast variances:

$$\sigma^2_{1t+1} = \mathbb{E}(e_{1t+1}|\Omega_t)^2 = \mathbb{E}(\varepsilon^2_{1t+1}) = \sigma^2_1$$
$$\sigma^2_{2t+1} = \mathbb{E}(e_{2t+1}|\Omega_t)^2 = \mathbb{E}(\varepsilon^2_{2t+1}) = \sigma^2_2$$

Assuming normality of forecast errors, interval forecasts are obtained the usual way.

Forecasting with VAR models: multi-step ahead

Point forecasts:

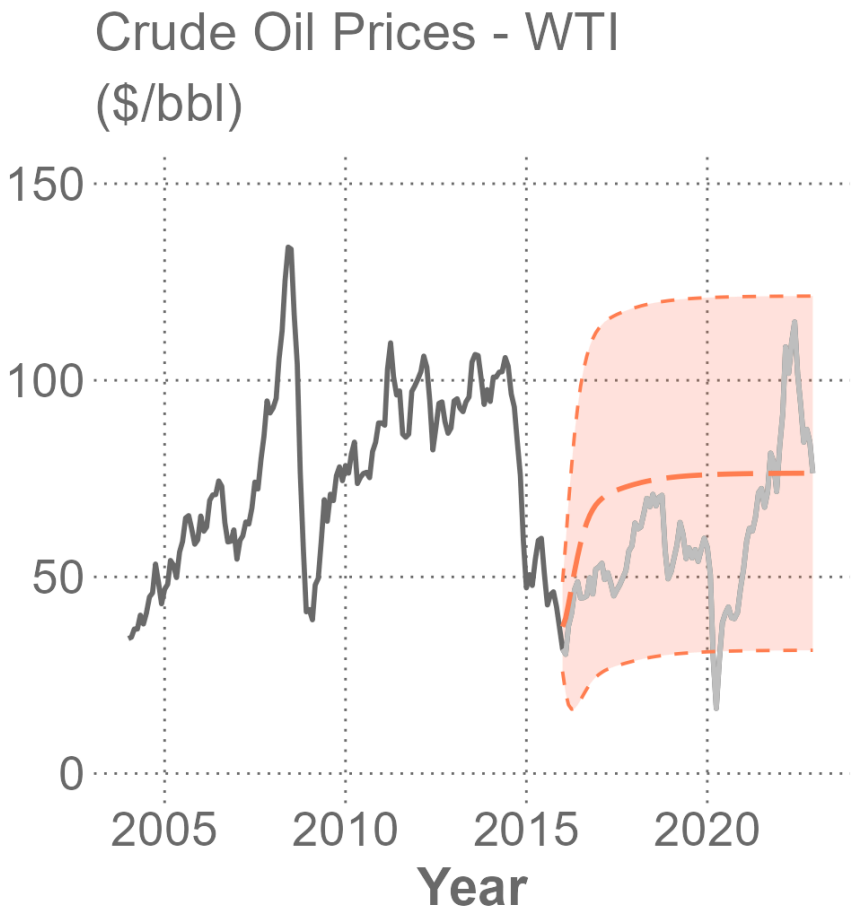
$$x_{1t+h|t} = \alpha_1 + \pi_{111} x_{1t+h-1|t} + \pi_{121} x_{2t+h-1|t}$$
$$x_{2t+h|t} = \alpha_2 + \pi_{211} x_{1t+h-1|t} + \pi_{221} x_{2t+h-1|t}$$

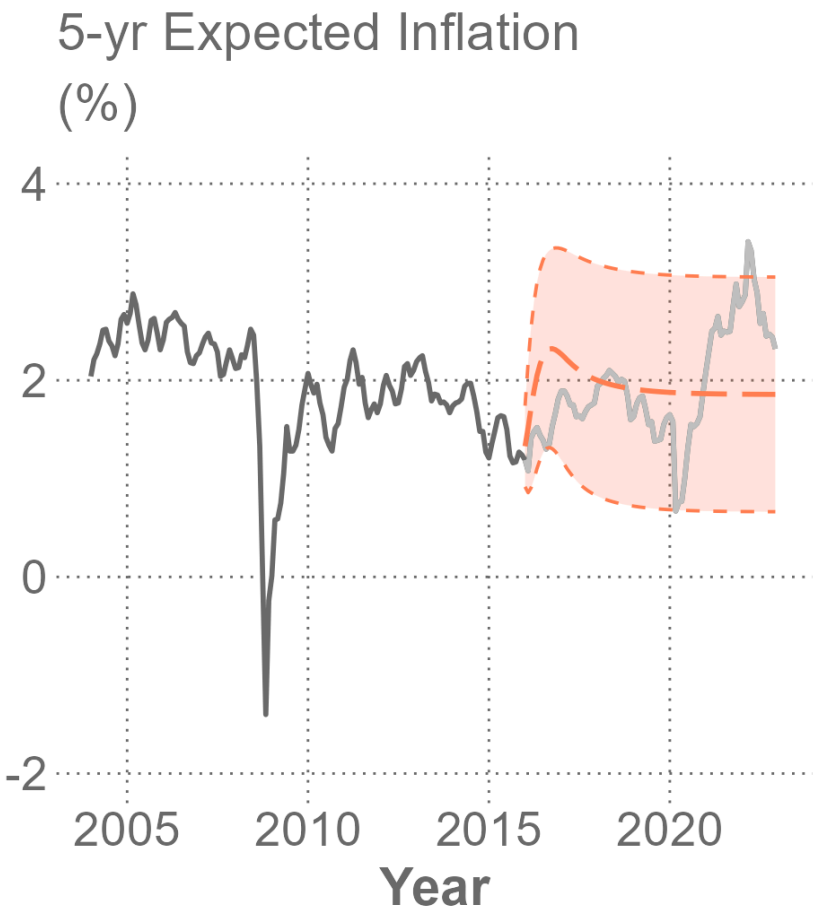
Forecast errors:

$$e_{1t+h} = \pi_{111} e_{1t+h-1} + \pi_{121} e_{2t+h-1} + \varepsilon_{1t+h}$$
$$e_{2t+h} = \pi_{211} e_{1t+h-1} + \pi_{221} e_{2t+h-1} + \varepsilon_{2t+h}$$

Forecast variances are the functions of error variances and covariances, and the model parameters.

Forecasting oil prices and inflation





Direct multi-step ahead forecasts

For a given horizon, h, the estimated model:

$$x_{1t} = \phi_1 + \psi_{111} x_{1t-h} + \psi_{121} x_{2t-h} + v_{1ht}$$
$$x_{2t} = \phi_2 + \psi_{211} x_{1t-h} + \psi_{221} x_{2t-h} + v_{2ht}$$

where $v_{1ht} \sim iid(0, \sigma^2_{1h})$ and $v_{2ht} \sim iid(0, \sigma^2_{2h})$, and the two can be correlated, i.e., $Cov(v_{1ht}, v_{2ht}) \neq 0$.

Point forecasts:

$$x_{1t+h|t} = \phi_1 + \psi_{111} x_{1t} + \psi_{121} x_{2t}$$
$$x_{2t+h|t} = \phi_2 + \psi_{211} x_{1t} + \psi_{221} x_{2t}$$

Direct multi-step ahead forecasts

Forecast errors:

$$e_{1t+h} = v_{1t+h}$$
$$e_{2t+h} = v_{2t+h}$$

Forecast variances:

$$\sigma^2_{1t+h} = \mathbb{E}(e_{1t+h}|\Omega_t)^2 = \mathbb{E}(v^2_{1t+h}) = \sigma^2_{1h}$$
$$\sigma^2_{2t+h} = \mathbb{E}(e_{2t+h}|\Omega_t)^2 = \mathbb{E}(v^2_{2t+h}) = \sigma^2_{2h}$$

Assuming normality, interval forecasts are obtained directly from these variances, the usual way.

Out-of-Sample Granger Causality

The previously discussed (in sample) tests of causality in Granger sense are frequently performed in practice, but the 'true spirit' of such test is to assess the ability of a variable to help predict another variable in an out-of-sample setting.

Out-of-Sample Granger Causality

Consider restricted and unrestricted information sets:

$$\Omega_t^{(r)} \equiv \Omega_t(X_1) = \{x_{1,t}, x_{1,t-1}, \ldots\}$$
$$\Omega_t^{(u)} \equiv \Omega_t(X_1, X_2) = \{x_{1,t}, x_{1,t-1}, \ldots, x_{2,t}, x_{2,t-1}, \ldots\}$$

Following Granger's definition of causality: $\{X_2\}$ is said to cause $\{X_1\}$ if $\sigma^2_{x_1}\left(\Omega_t^{(u)}\right) < \sigma^2_{x_1}\left(\Omega_t^{(r)}\right)$, meaning that we can better predict X_1 using all available information on X_1 and X_2, rather than that on X_1 only.

Out-of-Sample Granger Causality

Let the forecasts based on each of the information sets be:

$$x_{1t+h|t}^{(r)} = E\left(x_{1t+h}|\Omega_t^{(r)}\right)$$
$$x_{1t+h|t}^{(u)} = E\left(x_{1t+h}|\Omega_t^{(u)}\right)$$

Out-of-Sample Granger Causality

For these forecasts, the corresponding forecast errors are:

$$e^{(r)}_{1t+h} = x_{1t+h} - x^{(r)}_{1t+h|t}$$
$$e^{(u)}_{1t+h} = x_{1t+h} - x^{(u)}_{1t+h|t}$$

The out-of-sample forecast errors are then evaluated by comparing the loss functions based on these forecasts errors.

The out of sample Granger causality test is, in effect, a test of relative forecast accuracy between the two models.

Accuracy Tests for Nested Models

In testing Out-of-sample Granger causality, we compare nested models.

For example, consider a bivariate VAR(1). A test of out-of-sample Granger causality involves comparing forecasts from these two models:

$$
\begin{aligned}
(A): \quad & x_{1t} = \beta_0 + \beta_1 x_{1t-1} + \beta_2 x_{2t-1} + v_t \\
(B): \quad & x_{1t} = \alpha_0 + \alpha_1 x_{1t-1} + \varepsilon_t
\end{aligned}
$$

Obviously, here model (B) is nested in model (A). Under the null of Granger non-causality, the disturbances of the two models are identical. This leads to 'issues' in the usual tests.

Accuracy Tests for Nested Models

There is a way to circumvent these issues, which involves an adjustment of the loss differential.

In particular, the loss differential now becomes:

$$d(e_{t+h,ij}) = e^2_{t+h,i} - e^2_{t+h,j} + (y_{t+h|t,i} - y_{t+h|t,j})^2,$$

where model i is nested in model j.

The testing procedure is otherwise similar to the Diebold-Mariano test.

Key Takeaways

- Many economic time series are inter-related. We can model such series as a system of equations known as the Vector Autoregressive (VAR) model.
- Autoregressive Distributed Lag (ARDL) model is a special case (which is, in effect, a single equation) of a VAR model.
- We can use information on related variables to help better forecast a variable of interest. If improved accuracy is achieved, we have a case of Granger causality.